

Introduction

We will study the theorems related with Area.

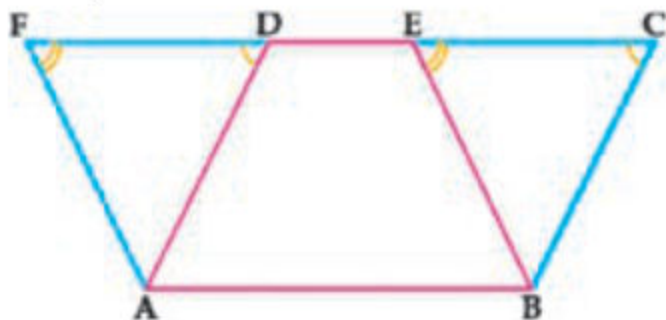
14.1 Theorems Related with Area

Theorem 14.1.1

Parallelograms on the same base and lying between the same parallel lines (or of the same altitude) are equal in area.

Given:

Two parallelograms ABCD and ABEF with the same base $\overline{AB}$ and between the same parallels segments $\overline{AB}$ and $\overline{DE}$.



To prove:

Parallelograms ABCD and ABEF are equal in areas,
i.e. $\blacksquare ABCD = \blacksquare ABEF$.

Proof:

Statements	Reasons
In $\triangle BCE \leftrightarrow \triangle ADF$	
(i) $m\overline{BC} = m\overline{AD}$... (i)	(i) Opposite sides of $\parallel^m$ ABCD are equal.
(ii) $m\angle BCE = m\angle ADF$... (ii)	(ii) Corresponding angles of $\parallel^m$ ABCD.
(iii) $\angle E \cong \angle F$... (iii)	(iii) Corresponding angles of $\parallel^m$ ABEF.
$\therefore \triangle BCE \cong \triangle ADF$	S.A.A $\cong$ S.A.A
$\therefore \triangle BCE \cong \triangle ADF$	Congruent figures are equal in area.
$\blacksquare ABED + \triangle BCE = \blacksquare ABED + \triangle ADF$	Adding same area on both sides
Thus, $\blacksquare ABCD = \blacksquare ABEF$.	$\blacksquare ABCD = \blacksquare ABED + \triangle BCE$ $\blacksquare ABEF = \blacksquare ABED + \triangle ADF$

Q.E.D

Corollary

- (i) The area of parallelogram is equal to that of a rectangle on the same base and having the same altitude.



Theorem 14.1.2

Parallelograms on equal bases and having the same altitude are equal in area.

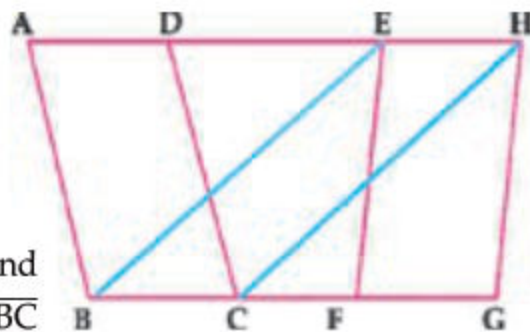
Given:

Parallelograms $ABCD$ and $EFGH$ are on the equal bases $\overline{BC}$ and $\overline{FG}$, having equal altitudes.

To Prove: $\blacksquare ABCD = \blacksquare EFGH$.

Construction:

Place the parallelograms $ABCD$ and $EFGH$ so that their equal bases $\overline{BC}$ and $\overline{FG}$ are on the same straight line. Join B to E and C to H .



Proof:

Statements	Reasons
$\parallel^m ABCD$ and $\parallel^m EFGH$ are between the same parallel segments $\overline{AH}$ and $\overline{BG}$. Hence, A, D, E and H are points lying on a straight line parallel to $\overline{BC}$.	Their altitudes are equal (given)
$m\overline{BC} = m\overline{FG}$	Given
$m\overline{BC} = m\overline{EH}$	$EFGH$ is a parallelogram and $m\overline{BC} = m\overline{FG}$
$m\overline{BC} = m\overline{EH}$ also these are parallel	Segment of parallel lines are also parallel segments.
Hence, $EBCH$ is a parallelogram	A quadrilateral with two parallel opposite sides is a parallelogram
Now $\blacksquare ABCD = \blacksquare EBCH$... (i)	Theorem 14.1.1
But $\blacksquare EBCH = \blacksquare EFGH$... (ii)	Theorem 14.1.1
Thus, $\blacksquare ABCD = \blacksquare EFGH$	From (i) and (ii)

Q.E.D



Theorem 14.1.3

Triangles on the same base and of the same altitude are equal in area.

Given:

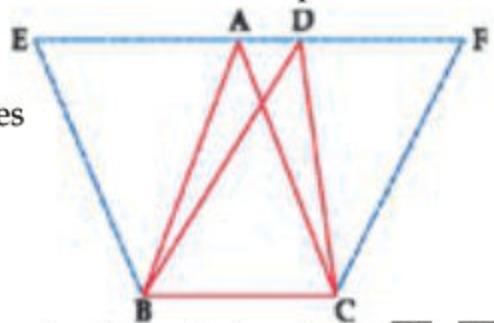
$\triangle ABC$ and $\triangle DBC$ are on the same base $\overline{BC}$ and between the same parallel lines $\overline{BC}$ and $\overline{AD}$.

To prove:

$$\triangle ABC = \triangle DBC$$

Construction:

Draw $\overline{BE} \parallel \overline{CA}$, meeting at $\overline{AD}$ produced, at E and also draw $\overline{CF} \parallel \overline{BD}$ meeting at $\overline{AD}$ produced at F.



Proof :

Statements	Reasons
BCAE is a parallelogram.	By construction
$\triangle ABC = \frac{1}{2} (\square BCAE)$... (i)	Diagonal $\overline{AD}$ divides parallelogram BCAE into two $\triangle$ s of equal areas.
Similarly BCDF is a parallelogram	By construction
$\triangle DBC = \frac{1}{2} (\square BCDF)$... (ii)	Diagonal $\overline{CD}$ divides parallelogram BCDF into two triangles of equal areas.
$\square BCAE = \square BCDF$... (iii)	Theorem 14.1.1
$\triangle ABC = \triangle DBC$	From (i),(ii) and (iii)

Q.E.D

Theorem 14.1.4

Triangles on equal bases and of equal altitudes are equal in area.

Given:

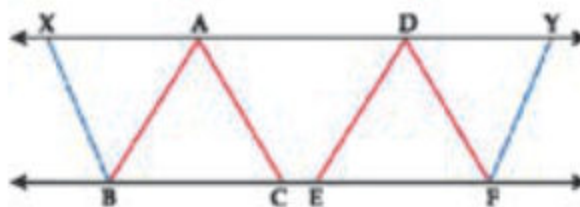
$\triangle ABC$ and $\triangle DEF$ are on equal bases $\overline{BC}$ and $\overline{EF}$ respectively and having equal altitudes.

To prove:

$$\triangle ABC = \triangle DEF$$

Construction:

Draw $\overline{AD}$, $\overline{BF}$ containing points B, C, E, F.





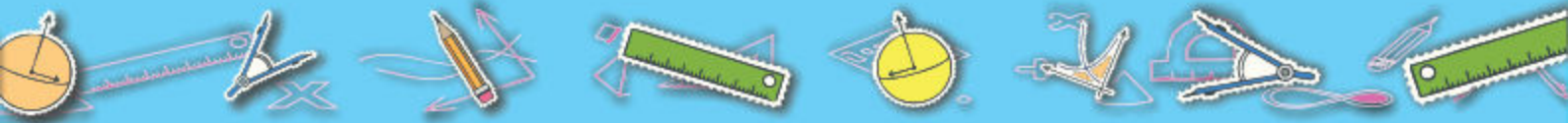
Place the $\triangle ABC$ and $\triangle DEF$ so that their equal bases $\overline{BC}$ and $\overline{EF}$ are on the straight line. Draw $\overline{BX} \parallel \overline{CA}$ and $\overline{FY} \parallel \overline{ED}$. Such that point X and Y lie on $\overline{AD}$.

Proof :

Statements	Reasons
$\triangle ABC$ and $\triangle DEF$ are between the same parallel lines.	Altitudes are equal (given)
$\overleftrightarrow{BF} \parallel \overleftrightarrow{XY}$	construction
$\therefore \blacksquare BCAX = \blacksquare EFYD \quad \dots (i)$	Theorem 14.1.2
But, $\triangle ABC = \frac{1}{2} (\blacksquare BCAX) \quad \dots (ii)$	Diagonal of a parallelogram divides $\parallel^m$ into two equal triangles
and $\triangle DEF = \frac{1}{2} (\blacksquare EFYD) \quad \dots (iii)$	By same reason
$\therefore \triangle ABC = \triangle DEF.$	From eqs.(i), (ii) and (iii)

Q.E.D

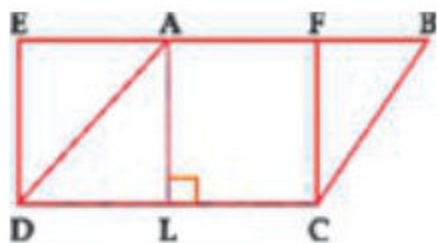
Corollary: Triangles having a common vertex and equal bases in the same straight line are equal in area.



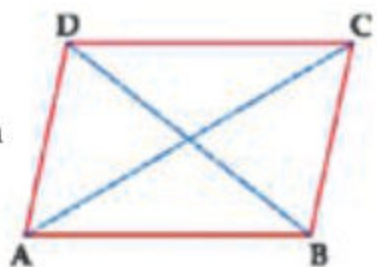
Exercise 14.1

1. In the given figure, $ABCD$ is a parallelogram and $EFCD$ is a rectangle, also $\overline{AL} \perp \overline{DC}$. Prove that

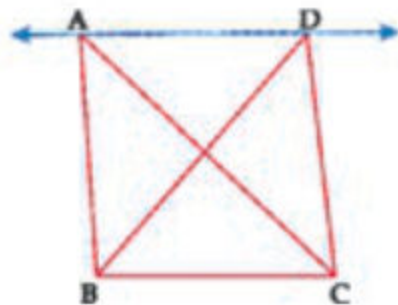
- (i) $\text{Area } ABCD = \text{Area } EFCD$
 (ii) $\text{Area } ABCD = m\overline{DC} \times m\overline{AL}$.



2. In the given figure, if the diagonals of a quadrilateral separate it into four triangles of equal area, show that it is a parallelogram.



3. In the given figure $\overline{BC} \parallel \overline{AD}$. $\triangle ABC$ is a right-angled at vertex B with $m\overline{BC} = 7$ cm and $m\overline{AC} = 11$ cm, also $\triangle ABC$ and $\triangle BCD$ are on the same base $\overline{BC}$. Find the area of $\triangle BCD$.



4. Show that a median of a triangle divides it into two triangles of equal area.
5. Show that the line segment joining the mid-points of the opposite sides of a rectangle, divides it into two equal rectangles.
6. If two parallelograms of equal areas have the same or equal bases, their altitudes are equal.
7. Show that an angle bisector of an equilateral triangle divides it into two triangles of equal areas.
8. Prove that a rhombus is divided by its diagonals into four triangles of equal areas.



Review Exercise 14

1. Mark 'T' for True and 'F' for False in front of each given below:

- (i) Area of a closed figure means region enclosed by bounding lines of the figure. T/F
- (ii) A diagonal of rectangle divides it into two congruent triangles. T/F
- (iii) Congruent figures have different areas. T/F
- (iv) The area of parallelogram is equal to the product of base and height. T/F
- (v) Median of a triangle means perpendicular from a vertex to the opposite side (base). T/F
- (vi) Perpendicular distance between two parallel lines can sometimes be different. T/F
- (vii) An altitude drawn from a vertex always bisects the opposite side. T/F
- (viii) Two triangles are equal in areas, if they have the same base and equal altitude. T/F

2. Tick (✓) the correct answer.

- (i) If perpendicular distance between two lines is the same. The lines are _____.

(a) Perpendicular to each other	(b) Parallel to each other
(c) Intersecting to each other	(d) None of these.
- (ii) If two triangles have equal areas then they will _____ be congruent as well.

(a) Not necessarily	(b) Necessarily
(c) Definitely	(d) None of these.
- (iii) Perpendicular from a vertex of a triangle to its opposite side is called _____.

(a) Median	(b) Perpendicular bisector
(c) Altitude	(d) Angle bisector



- (iv) Parallelograms having same base and same altitude are _____.
 (a) Congruent (b) Equal in areas
 (c) Similar (d) All of these.
- (v) Two parallelograms have equal bases. They will be having the same area, if _____.
 (a) Their altitudes are equal
 (b) Their altitude is the same
 (c) They lie between the same parallel lines
 (d) All of these.
- (vi) $\triangle ABC$ and $\triangle DEF$ have equal bases and equal altitudes, then triangles are _____.
 (a) Equal in area (b) Congruent
 (c) Similar (d) None of these.



Summary

- ◆ In this unit we have mentioned some necessary preliminaries, stated and proved the following theorems along with corollaries, if any.
- ◆ Parallelograms on the same base and between the same parallel lines (or of the same altitude) are equal in areas.
- ◆ Parallelograms on the equal bases and having the same (or equal) altitude are equal in areas.
- ◆ Triangles on the same base and of the same (i.e. equal) altitudes are equal in areas.
- ◆ Triangles on equal bases and of equal altitudes are equal in areas.
- ◆ Area of a figure means region enclosed by the boundary lines of a closed figure.
- ◆ A triangular region means the union of triangle and its interior.
- ◆ By area of triangle means the area of its triangular region.
- ◆ Altitude or height of a triangle means perpendicular distance to base from its opposite vertex.

Unit

15

• Weightage = 7%

PROJECTION OF A SIDE OF A TRIANGLE

Student Learning Outcomes (SLOs)

After completing this unit, students will be able to:

- ◆ In an obtuse-angled triangle, the square on the side opposite to the obtuse angle is equal to the sum of the squares on the sides containing the obtuse angle together with twice the rectangle contained by one of the sides, and the projection on it of the other.
- ◆ In any triangle, the square on the side opposite to an acute angle is equal to the sum of the squares on the sides containing that acute angle diminished by twice the rectangle contained by one of those sides and the projection on it of the other.
- ◆ In any triangle, the sum of the squares on any two sides is equal to twice the square on half of the third side together with twice the square on the median which bisects the third side, (Apollonius' theorem).